

BASIC SIGHT-SINGING AND EAR TRAINING FOR CHURCH SINGERS

UNIT 9.2 – PITCH BASIC HARMONY –

THE ANATOMY OF TRIADS IN 4-PART CHORAL WRITING. INVERSIONS.



HARMONY IN FOUR-PART CHORAL WRITING

In Unit 8.2 we introduced **triads**—chords of three notes constructed from the various pitches of the major and minor scales. When triads are sung by a choir, the pitches can be **distributed** or **assigned** among the different choral voices in various ways.

A four-part choir can be one of two types:

1. a **mixed** choir—consisting of women's (or children's) voices (Sopranos and Altos) and men's voices (Tenors and Basses); or
2. a **single-gender** choir—consisting of only women's voices (Soprano 1, Soprano 2, Alto 1, Alto 2) or only men's voices (Tenor 1, Tenor 2, Bass 1, Bass 2).

THE NAMES OF CHORAL VOICE PARTS

The names of the voice parts have their origins in the late Middle Ages and early Renaissance (14th-15th centuries), when singing in parts first developed in Western Europe. Music of that time was frequently based on melodies of Gregorian Chant; the chant melody was assigned to be “held” by the **tenor**, from the Latin word *tenere*—to hold. The part below the tenor was called the **contratenor bassus**, while the part above the tenor was called the **contratenor altus**. The part that sang above all the other parts was called the **superius**. Eventually, the “contratenor” was dropped from the name, and we got our standard names of the voice parts:

Soprano

Alto

Tenor

Bass

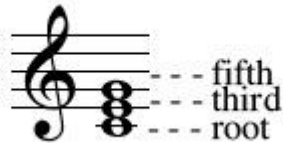
THE ANATOMY OF A TRIAD

As mentioned in Unit 8.2, a triad by definition consists of **three pitches**, stacked in intervals of thirds on top of a starting note.

The starting or bottom note is referred to as the **root** of the triad or, more generally, the **root of the chord**.

The second component note of a triad, which is a third above the root, is called the **third of the chord**.

The third note, a fifth above the root is called, predictably, the **fifth of the chord**.
For example:



Shown in this manner, these names—root, third, and fifth—are somewhat abstract, because, as we shall see below, they only describe the **conceptual** aspect of a chord—its component notes, but don't yet tell us anything specific about how a given chord is constructed or how it is **voiced**—how are the pitches of the triad distributed among the various voice parts of the choir.

Because there are only three notes in the triad, but four choral parts, there are different ways of distributing the notes of the chord, different **voicings**. One of the parts will be, of necessity, doubled.

Here are four different possible voicings for the C Major triad, when sung by a mixed choir:

The image displays four musical staves, each representing a different voicing of the C Major triad (C4, E4, G4) for a mixed choir. The staves are labeled Soprano, Alto, Tenor, and Bass. The notes are represented by whole notes. The voicings are as follows:

- Staff 1:** Soprano (C4), Alto (E4), Tenor (G4), Bass (C4). Labels: Soprano --- root, Alto --- fifth, Tenor --- third, Bass --- root.
- Staff 2:** Soprano (C4), Alto (C4), Tenor (E4), Bass (G4). Labels: Soprano --- root, Alto --- third, Tenor --- fifth, Bass --- root.
- Staff 3:** Soprano (E4), Alto (G4), Tenor (C4), Bass (C4). Labels: Soprano --- fifth, Alto --- third, Tenor --- root, Bass --- root.
- Staff 4:** Soprano (E4), Alto (C4), Tenor (G4), Bass (C4). Labels: Soprano --- third, Alto --- root, Tenor --- fifth, Bass --- root.

A similar array of possible voicings exists for the F Major triad.

Four musical staves showing different voicings of the F Major triad (F-A-C) in root position. Each staff shows Soprano, Alto, Tenor, and Bass parts. The first staff shows the root (F) in the bass and the fifth (C) in the soprano. The second staff shows the root (F) in the bass and the third (A) in the soprano. The third staff shows the root (F) in the bass and the fifth (C) in the soprano. The fourth staff shows the root (F) in the bass and the fifth (C) in the soprano.

And here are some different voicings of the E minor triad.

Four musical staves showing different voicings of the E minor triad (E-G-B) in root position. Each staff shows Soprano, Alto, Tenor, and Bass parts. The first staff shows the root (E) in the bass and the fifth (B) in the soprano. The second staff shows the root (E) in the bass and the third (G) in the soprano. The third staff shows the root (E) in the bass and the fifth (B) in the soprano. The fourth staff shows the root (E) in the bass and the fifth (B) in the soprano.

Because in each of the triads above **the root of the chord is on the bottom**, these triads are said to be in **root position**.

We note that every triad on the previous two slides has a **root**, **third**, and **fifth**, and the root is **doubled**, that is, it is assigned to more than one voice part. There is a reason for this, grounded in physics and acoustics: this type of chord sounds most **stable** acoustically, is easiest to **tune**, and has been shown to be most pleasant-sounding to the ear because of the interplay of the **overtones** that are generated when any pitch is sounded.

There are other possible voicings, e.g., with the **third doubled** or the **fifth doubled**, but these chords tend not to be as stable and are more difficult to tune. Consequently, theory and composition textbooks caution composers against extensive use of such triads in writing simple chords for four-part choir.

CHORD INVERSIONS

It is possible to build chords with notes other than the root on the bottom. When the bottom note is the third of the chord, it is said to be in **first inversion**.

The other parts of the triad (the root and the fifth) may be distributed in various ways above the third, and the third itself may be doubled in another voice.

Here are some different possible voicings for the C Major chord in first inversion.

Three musical staves illustrating different voicings for the C Major chord in first inversion. The staves are labeled Soprano, Alto, Tenor, and Bass. The notes are C, E, and G. The first staff shows the root (C) in the Soprano and Alto voices, the fifth (G) in the Tenor voice, and the third (E) in the Bass voice. The second staff shows the root (C) in the Soprano voice, the fifth (G) in the Alto and Tenor voices, and the third (E) in the Bass voice. The third staff shows the fifth (G) in the Soprano voice, the root (C) in the Alto voice, the third (E) in the Tenor voice, and the fifth (G) in the Bass voice.

Here are some examples of the G Major triad in first inversion.

Three musical examples of the G Major triad in first inversion, showing the distribution of notes (B, D, F#) across four voices (Soprano, Alto, Tenor, Bass) in first inversion. The notes are labeled as root, fifth, and third.

- Example 1:** Soprano (B, root), Alto (D, fifth), Tenor (F#, third), Bass (B, root).
- Example 2:** Soprano (D, fifth), Alto (F#, third), Tenor (B, root), Bass (D, fifth).
- Example 3:** Soprano (F#, third), Alto (B, root), Tenor (D, fifth), Bass (F#, third).

And here are some examples of A minor triad in first inversion.

Three musical examples of the A minor triad in first inversion, showing the distribution of notes (C, E, G) across four voices (Soprano, Alto, Tenor, Bass) in first inversion. The notes are labeled as fifth, root, and third.

- Example 1:** Soprano (C, fifth), Alto (E, root), Tenor (G, third), Bass (C, fifth).
- Example 2:** Soprano (E, root), Alto (G, third), Tenor (C, fifth), Bass (E, root).
- Example 3:** Soprano (G, third), Alto (C, fifth), Tenor (E, root), Bass (G, third).

When the bottom note is the **fifth** of the chord, it is said to be in **second inversion**. As in the case of root-position and first-inversion triads, the other parts of the triad (the root and the third) may be distributed in various ways above the fifth. Second-inversion triads, however, are considered **acoustically unstable** and are therefore encountered rarely in choral music that is written in accordance with the rules of harmony and part-writing. They do occasionally occur in passing, particularly in chord progressions that culminate on the tonic chord (called **cadential chord progressions** or **cadences** for short), as shown below.



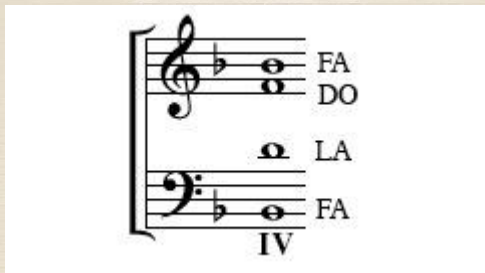
TRIADS, INVERSIONS, AND SOLFÈGE

It is possible to apply SOL-FA syllables to the notes of triads and their inversions, as long as one knows what the **key** is, and what note is the **major tonic**—DO, or **minor tonic**—LA. Thus, in a major key, the triad in root position built upon the **I** degree—DO, would be spelled DO-MI-SOL and called “**the I chord**”; upon the **ii** degree—RE, would be RE-FA-LA and called “**the ii chord**”; upon the **iii** degree—MI, would be MI-SOL-TI, called “**the iii chord**,” and so forth.

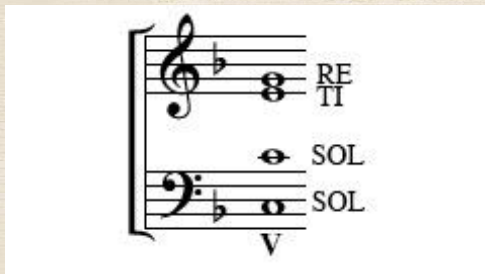
For example, for a triad built on the **I** degree in F major, the root would be DO, the third would be MI, and the fifth would be SOL, no matter what order those pitches occur from the bottom notes of the chord to the top note of the chord.



The component parts of the triad built upon the **IV** degree (called the **subdominant**) are: the root—FA, the third—LA, and the fifth—DO:



The component parts of the triad built upon the **V** degree (called the **dominant**) are: the root—SOL, the third—TI, and the fifth—RE:



Note that the **same** SOL-FA names are applied to the component notes of the triad if it is in first or second inversion. For example, in F major:



It is possible to examine all four voices of a choral piece from the standpoint of solfège and, simultaneously, from the standpoint of the triads that are formed as the voices move from beginning to end. Seen **horizontally**, each voice spells out a melody that can be expressed in SOL-FA syllables. At the same time, viewed **vertically**, each chord can also be spelled out in SOL-FA syllables. These two dimensions represent the marvelous interplay of melody and harmony that gives four-part choral music its character and beauty. (See following slide)

Soprano
Alto

DO DO RE MI FA RE MI RE DO

SOL SOL TI DO TI TI DO

Re - ceive the Bod - y of Christ;

MI MI SOL SOL LA FA SOL FA MI

Tenor
Bass

DO DO SOL DO FA SOL SOL LA

ON THE SPOT EXERCISE:

Using the example above, find the tonic F=DO. Sing **each voice**—Soprano, Alto, Tenor, and Bass, using SOL-FA syllables. Then sing **each chord** vertically, using SOL-FA syllables: sing the pitches either from the top down, or—better, by first naming each voice part and the note it is singing, for example, Soprano-DO, Alto-SOL, Tenor-MI, Bass-DO.

Please proceed to **PITCH EXERCISES 9.2**

